

INTRODUCTION TO LINUX FORENSICS CHEAT SHEET

Command	Description
<code>tail -n 10 /var/log/syslog</code>	View recent syslog entries on a Debian-based system
<code>dmesg</code>	Check kernel logs
<code>python3 zircolite.py --events audit.log --auditd --ruleset rules/rules_linux.json --csv -o audit_logs_zircolite.csv</code>	Parse auditd logs using SIGMA rules with Zircolite and export detections to CSV
<code>aureport -if audit.log</code>	Generate interpreted file access report from audit logs
<code>ausearch --syscall EXECVE -if audit.log</code>	Search audit logs for process execution events (EXECVE syscalls) in interpreted format

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<code>ausearch -if audit.log -m ADD_USER -m DEL_USER -m ADD_GROUP -m USER_CHAUTHOK -m DEL_GROUP -m CHGRP_ID -m ROLE_ASSIGN -m ROLE_REMOVE -i</code>	Search audit logs for user and group management events (add/delete users/groups, role changes)
<code>sudo auditctl -s</code>	Display the current status of the audit daemon
<code>stat /etc/sudoers</code>	Display metadata (timestamps, permissions) of the sudoers file
<code>python3 ~/volatility3/vol.py -q -f /path/to/server-dump linux.bash.Bash</code>	Recover bash command history from memory dump
<code>python3 ~/volatility3/vol.py -q -f /path/to/server-dump banners.Banners</code>	Identify Linux kernel version and OS banners from memory dump
<code>python3 ~/volatility3/vol.py -q -f /path/to/server-dump linux.sockstat.Sockstat</code>	List active network sockets and connections from memory
<code>python3 ~/volatility3/vol.py -q -f /path/to/server-dump linux.pslist</code>	Enumerate running processes from memory dump
<code>python3 ~/volatility3/vol.py -q -f /path/to/server-dump linux.netstat</code>	Display network connections and ports from memory
<code>python3 ~/volatility3/vol.py -q -f /path/to/server-dump linux.pstree</code>	Display process parent-child relationships (process tree) from memory

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<code>python3 ~/volatility3/vol.py -q -f /path/to/server-dump linux.lsmmod</code>	Enumerate loaded kernel modules from memory dump
<code>python3 ~/volatility3/vol.py -q -f /path/to/server-dump linux.lsof</code>	List files opened by processes from memory
<code>python3 ~/volatility3/vol.py -q -f /path/to/server-dump linux.check_syscall</code>	Detect modifications to the syscall table from memory
<code>python3 ~/volatility3/vol.py -q -f /path/to/server-dump linux.elfcore</code>	Scan and recover ELF executables from memory dump
<code>python3 ~/volatility3/vol.py -q -f /path/to/server-dump linux.envvars --pid 28645</code>	Extract environment variables for a specific process from memory
<code>python3 ~/volatility3/vol.py -q -f /path/to/server-dump linux.malfind.Malfind --pid 28645</code>	Scan for hidden or injected code in process memory
<code>python3 ~/volatility3/vol.py -q -f /path/to/server-dump linux.proc.Maps --dump --pid 910</code>	Dump virtual memory mappings for a process to file
<code>strings -a pid.1441.vma.0x7fa7fc0c0000-0x7fa7fc0c6000.dmp</code>	Extract printable strings from a dumped memory segment
<code>sudo ./avml memdump.mem</code>	Acquire Linux volatile memory dump using AVML tool
<code>journalctl --utc</code>	Display timestamps in UTC

Command	Description
<code>journalctl -b</code>	Display logs from current boot
<code>journalctl --list-boots</code>	List previous boots
<code>journalctl -b -1</code>	See journal from the previous boot (you can use boot ID instead -1)
<code>journalctl --utc -D /var/log/journal/894062f9af204645a289e8016977fe6c</code>	Use external journal folder to retrieve results from
<code>journalctl --utc --since "2023-10-15 18:00:00" --until "2023-10-15 19:00:00" -D /var/log/journal/894062f9af204645a289e8016977fe6c</code>	Use time windows for your logs (format: YYYY-MM-DD HH:MM:SS)
<code>journalctl --utc --since "2023-10-15 18:00:00" --until "2023-10-15 19:00:00" -D /var/log/journal/894062f9af204645a289e8016977fe6c/ -u httpd.service</code>	Filter results by unit (httpd.service as an example)
<code>journalctl --utc --since "2023-10-15 18:00:00" --until "2023-10-15 19:00:00" -D /var/log/journal/894062f9af204645a289e8016977fe6c/ -u httpd.service _PID=27804</code>	Filter results by Process ID (PID)
<code>journalctl /usr/bin/bash</code>	Filter results by executable
<code>journalctl -p err --no-pager</code>	Filter results by priority
<code>journalctl -o json-pretty</code>	Output journalctl in json or any other formats